

The performance of the *3LLR* algorithm has been studied and compared with the four aforementioned algorithms and the best solutions found (*BKS*). Table 1 and Table 2 compare the results of solving the literature algorithms and the *3LLR* algorithm for the first and second sets of instances, respectively.

Table 1: Percentage of gap of the *3LLR* algorithm compared to the 3 algorithms and the best solution found in the first set of instances

Name instance	Objective function value					gap percentage of <i>3LLR</i> algorithm compared to			
	<i>BKS</i>	<i>GRASP</i>	<i>MAPM</i>	<i>LRGTS</i>	<i>3LLR</i>	<i>BKS</i>	<i>GRASP</i>	<i>MAPM</i>	<i>LRGTS</i>
20-5-1	54793	55021	54793	55131	54793	0	-0.41	0	-0.61
20-5-1b	39104	39104	39104	39104	39104	0	0	0	0
20-5-2	48908	48908	48908	48908	48908	0	0	0	0
20-5-2b	37542	37542	37542	37542	37595	0.14	0.14	0.14	0.14
50-5-1	90111	90632	90160	90160	90459	0.39	-0.19	0.33	0.33
50-5-1b	63242	64761	63242	63256	64863	2.56	0.16	2.56	2.54
50-5-2	88298	88786	88298	88715	89886	1.8	1.24	1.8	1.32
50-5-2b	67308	68042	67893	67698	69027	2.55	1.45	1.67	1.96
50-5-2bis	84055	84055	84055	84148	84769	0.85	0.85	0.85	0.7
50-5-2bbis	51822	52059	51822	51992	51903	0.16	-0.30	0.16	-0.17
50-5-3	86203	87380	86203	86203	87964	2.04	0.67	2.04	2.04
50-5-3b	61830	61890	61830	61830	63409	2.55	2.45	2.55	2.55
100-5-1	274814	279437	281944	277935	281705	2.51	0.81	-0.08	1.36
100-5-1b	213598	216159	216656	214885	219994	2.99	1.77	1.54	2.38
100-5-2	193671	199520	195568	196545	198402	2.44	-0.56	1.45	0.94
100-5-2b	157095	159550	157325	157792	161526	2.82	1.24	2.67	2.37
100-5-3	200079	203999	201749	201952	202231	1.08	-0.87	0.24	0.14
100-5-3b	152441	154596	153322	154709	159311	4.51	3.05	3.91	2.97
100-10-1	287661	323171	316575	291887	295426	2.7	-8.59	-6.68	1.21
100-10-1b	230989	271477	270025	235532	243497	5.41	-10.31	-9.82	3.38
100-10-2	243590	254087	245123	246708	251013	3.05	-1.21	2.4	1.74
100-10-2b	203988	206555	205052	204435	211089	3.48	2.2	2.94	3.25
100-10-3	250882	270826	253669	258656	256172	2.11	-5.41	0.99	-0.96
100-10-3b	203114	216173	204815	205883	210378	3.58	-2.68	2.72	2.18
200-10-1	474850	490820	483497	481676	491078	3.42	0.05	1.57	1.95
200-10-1b	375177	416753	380044	380613	391547	4.36	-6.05	3.03	2.87
200-10-2	448077	512679	451840	453353	458965	2.43	-10.48	1.58	1.24
200-10-2b	373696	379980	375019	377351	381922	2.2	0.51	1.84	1.21
200-10-3	469433	496694	478132	476684	480678	2.4	-3.22	0.53	0.84
200-10-3b	362320	389016	364834	365250	373858	3.18	-3.90	2.47	2.36
Average	196289.7	207322.4	200301.3	198552.2	201715.73	2.26	-1.25	0.85	1.41

Table 2: Percentage gap of the 3LLR algorithm compared to the 4 algorithms and the best solution found in the second set of instances

Name instance	Objective function value						gap percentage of 3LLR algorithm compared to				
	BKS	Tuzun & Burke	GRASP	MAPM	LRGTS	3LLR	BKS	Tuzun & Burke	GRASP	MAPM	LRGTS
111112	1467.68	1556.64	1525.25	1493.92	1490.82	1527.81	4.1	-1.85	0.17	2.27	2.48
111122	1448.37	1553.88	1526.9	1471.36	1471.76	1515.45	4.63	-2.47	-0.75	3	2.97
111212	1394.8	1443.43	1423.54	1418.83	1412.04	1458.75	4.58	1.06	2.47	2.81	3.31
111222	1432.29	1511.39	1482.29	1492.46	1443.06	1488.77	3.94	-1.50	0.44	-0.25	3.17
112112	1167.16	1231.11	1200.24	1173.22	1187.63	1198.48	2.68	-2.65	-0.15	2.15	0.91
112122	1102.24	1132.02	1123.64	1115.37	1115.95	1152.3	4.54	1.79	2.55	3.31	3.26
112212	791.66	825.12	814	793.97	813.28	823.42	4.01	-0.21	1.16	3.71	1.25
112222	728.3	740.54	747.84	730.51	742.96	751.45	3.18	1.47	0.48	2.87	1.14
113112	1238.49	1316.98	1273.1	1262.32	1263.97	1300.43	5	-1.26	2.15	3.02	2.88
113122	1245.3	1274.5	1272.94	1251.32	1256.12	1290.72	3.65	1.27	1.4	3.15	2.75
113212	902.26	920.75	912.19	903.82	913.06	917.03	1.64	-0.40	0.53	1.46	0.43
113222	1018.29	1042.21	1022.51	1022.93	1025.51	1045.24	2.65	0.29	2.22	2.18	1.92
121112	2237.73	2379.47	2384.01	2293.99	2296.52	2402.99	7.39	0.99	0.8	4.75	4.64
121122	2137.45	2211.74	2288.09	2277.39	2207.5	2285.34	6.92	3.33	-0.12	0.35	3.53
121212	2195.17	2288.17	2273.19	2274.57	2260.8	2324.93	5.91	1.61	2.28	2.21	2.83
121222	2214.86	2355.81	2345.1	2376.25	2259.25	2373.11	7.14	0.73	1.19	-0.13	5.04
122112	2070.43	2158.6	2137.08	2106.36	2120.76	2175.78	5.09	0.8	1.81	3.3	2.59
122122	1685.52	1787.02	1807.29	1771.53	1737.81	1791.52	6.29	0.25	-0.87	1.13	3.09
122212	1449.93	1549.79	1496.75	1467.54	1488.55	1533.09	5.74	-1.08	2.43	4.47	2.99
122222	1082.46	1112.96	1095.92	1088	1090.59	1113.64	2.88	0.06	1.62	2.36	2.11
123112	1942.23	2056.11	2044.66	1973.28	1984.06	2021.36	4.07	-1.69	-1.14	2.44	1.88
123122	1910.08	2002.42	2090.95	1979.05	1986.49	2029.63	6.26	1.36	-2.93	2.56	2.17
123212	1760.9	1877.3	1788.7	1782.23	1786.79	1836.34	4.28	-2.18	2.66	3.04	2.77
123222	1390.86	1414.83	1408.63	1396.24	1401.16	1427.16	2.61	0.87	1.32	2.21	1.86
131112	1892.17	2000.97	2006.7	1959.39	1946.01	2018.95	6.7	0.9	0.61	3.04	3.75
131122	1819.68	1892.84	1888.9	1881.67	1875.79	1930.82	6.11	2.01	2.22	2.61	2.93
131212	1986.02	2022.11	2033.93	1984.25	2010.53	2074.04	4.43	2.57	1.97	4.53	3.16
131222	1792.77	1854.92	1856.07	1855.25	1819.89	1905.43	6.28	2.72	2.66	2.7	4.7
132112	1443.32	1555.82	1508.32	1448.27	1448.65	1532.51	6.18	-1.50	1.6	5.82	5.79
132122	1429.3	1478.8	1456.82	1459.83	1492.86	1490.97	4.31	0.82	2.34	2.13	-0.13
132212	1204.42	1231.4	1240.4	1207.41	1211.07	1231.62	2.26	0.02	-0.71	2.01	1.7
132222	924.68	948.28	940.8	934.79	936.93	962.74	4.12	1.52	2.33	2.99	2.75
133112	1694.18	1762.45	1736.9	1720.3	1729.31	1763.12	4.07	0.04	1.51	2.49	1.96
133122	1392.01	1488.34	1425.74	1429.34	1424.54	1464.48	5.21	-1.60	2.72	2.46	2.8
133212	1197.95	1264.33	1223.7	1203.44	1216.32	1236.16	3.19	-2.25	1.02	2.72	1.63
133222	1151.37	1182.28	1231.33	1158.54	1162.12	1186.84	3.08	0.39	-3.61	2.44	2.13
Avrage	1498.4	1567.38	1556.51	1532.19	1528.63	1571.73	4.59	0.17	1.01	2.62	2.64

Thirteen small-scale instances are created to solve the 3L_LRM model. The instance information is available in Table 3, which includes the number of areas, depots, goods, and patterns.

Table 3: Informations for small-scale issues

instance	Number of			
	area	depot	good	pattern
NO.01	5	2	2	2
NO.02	5	2	3	3
NO.03	5	2	3	4
NO.04	7	2	2	2
NO.05	7	2	3	3
NO.06	7	2	3	4
NO.07	9	2	2	2
NO.08	9	2	3	3
NO.09	9	2	3	4
NO.10	12	2	2	2
NO.11	12	2	2	3
NO.12	15	2	2	2
NO.13	15	2	2	3

The dimensions and costs of using each vehicle type for the instances are shown in Table 4. The first and second sets of instances are run with two sets of patterns. The first set of patterns has 3 types of goods and 4 types of patterns, while the second set has 5 types of goods and 7 types of patterns. Information about the dimensions of the patterns is shown in Table 5 and Table 6.

Table 4: Dimensions and cost of each type of vehicle for the first and second sets

set	vehicle	length (cm)	width (cm)	height (cm)	cost
first	Type 1	587	233	220	1000
	Type 2	297	222	220	700
second	Type 1	587	233	220	10
	Type 2	297	222	220	7

Table 5: Dimensions and information of the first set of patterns

pattern	length (cm)	width (cm)	height (cm)	Number of good
Type 1	108	76	30	1 item of type 1
Type 2	110	43	25	1 item of type 2
Type 3	92	81	55	1 item of type 3
Type 4	110	86	25	2 item of type 2

Table 6: Dimensions and information of the second set of patterns

pattern	length (cm)	width (cm)	height (cm)	Number of good
Type 1	49	25	21	1 item of type 1
Type 2	60	51	41	1 item of type 2
Type 3	103	76	64	1 item of type 3
Type 4	95	70	62	1 item of type 4
Type 5	111	49	26	1 item of type 5
Type 6	49	50	21	2 item of type 1
Type 7	102	60	41	2 item of type 2

The number of walls, columns, and layers considered for each vehicle type is shown in Table 7. For each vehicle type, the average width of the goods types in each pattern set is calculated, and the vehicle's length and width are divided by this average; the resulting numbers are taken as the number of walls and the number of columns in each wall. For the number of layers in each column, the height of each type of vehicle is divided by the average height of the goods in each pattern set.

Table 7: The number of walls, columns, and layers of each vehicle

vehicle	wall	column	layer
Type 1- first set of patterns	9	4	7
Type 2- first set of patterns	5	4	7
Type 1- second set of patterns	11	5	6
Type 2- second set of patterns	6	5	6

There are five types of goods for distribution: blankets, dry goods, water, tents, and heating devices. To implement the *3LLR* algorithm in the case study, eleven pattern types for these five types of goods are considered, and the information on the dimensions of the patterns is shown in Table 8.

Table 8: Dimensions and information of 5 types of goods and 11 types of patterns for case study

Good name	pattern	length (cm)	width (cm)	height (cm)	Number of good
Blanket	Type 1	55	40	60	1 item of type 1
Dry food	Type 2	40	30	30	1 item of type 2
	Type 3	80	30	30	2 item of type 2
	Type 4	60	40	30	2 item of type 2
Water	Type 5	25	16	31	1 item of type 3
	Type 6	50	16	31	2 item of type 3
	Type 7	50	32	31	4 item of type 3
Tent	Type 8	100	40	33	1 item of type 4
	Type 9	100	80	33	2 item of type 4
Heating device	Type 10	60	30	40	1 item of type 5
	Type 11	60	60	40	2 item of type 5